HP-EGR CONTROL VALVE [SKYACTIV-D 1.5]

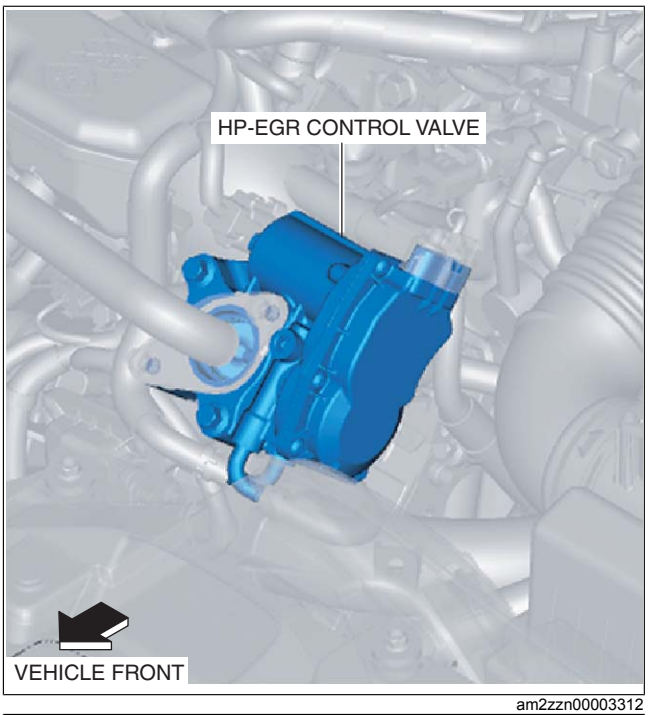
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Purpose, Function

- The HP-EGR control valve adjusts the exhaust gas volume recirculated to the combustion chamber according to the engine operation conditions to reduce the NOx generated largely when the combustion temperature in the combustion chamber is high.

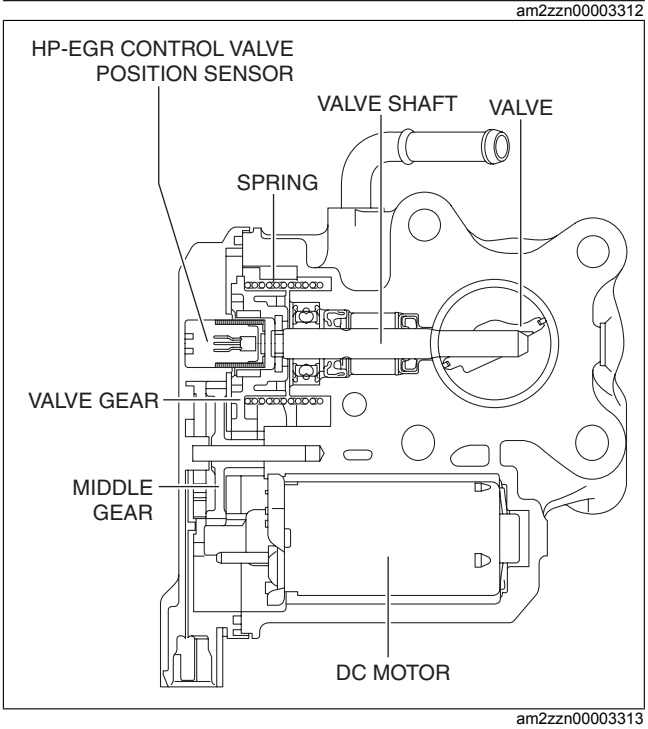
Construction

- The HP-EGR control valve is located at the front of the engine compartment.



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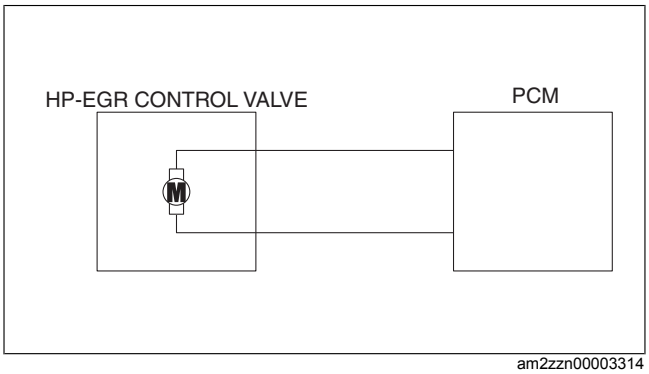
- The HP-EGR control valve consists of the DC motor, valve, spring, valve gear, and mid gear.
- The HP-EGR control valve has an integrated HP-EGR control valve position sensor. For details on the HP-EGR control valve position sensor, refer to the CONTROL SYSTEM. (See HP-EGR CONTROL VALVE POSITION SENSOR [SKYACTIV-D 1.5].)



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Operation

- The HP-EGR control valve is opened to the valve opening angle according to the energization current flowing to the DC motor from the PCM.
- The HP-EGR control valve performs coordinated control with the intake shutter valve, LP-EGR control valve, and exhaust shutter valve. For details on the HP-EGR control valve control, refer to the CONTROL SYSTEM. (See EGR CONTROL [SKYACTIV-D 1.5].)
- In the event that HP-EGR control valve has a malfunction and the DC motor control is disabled, the shutter valve is closed to the fully closed position by the return spring. As a result, the air volume necessary to keep the vehicle moving is assured.



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Fail-safe

DTC No.	Fail-safe function
P0401:00	• Inhibits engine-stop by operating the i-stop function.
P0402:00	• Limits the engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function.
P0404:00	• Limits the engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function. • PCM restricts engine-transaxle integration control. (ATX)
P0488:00	